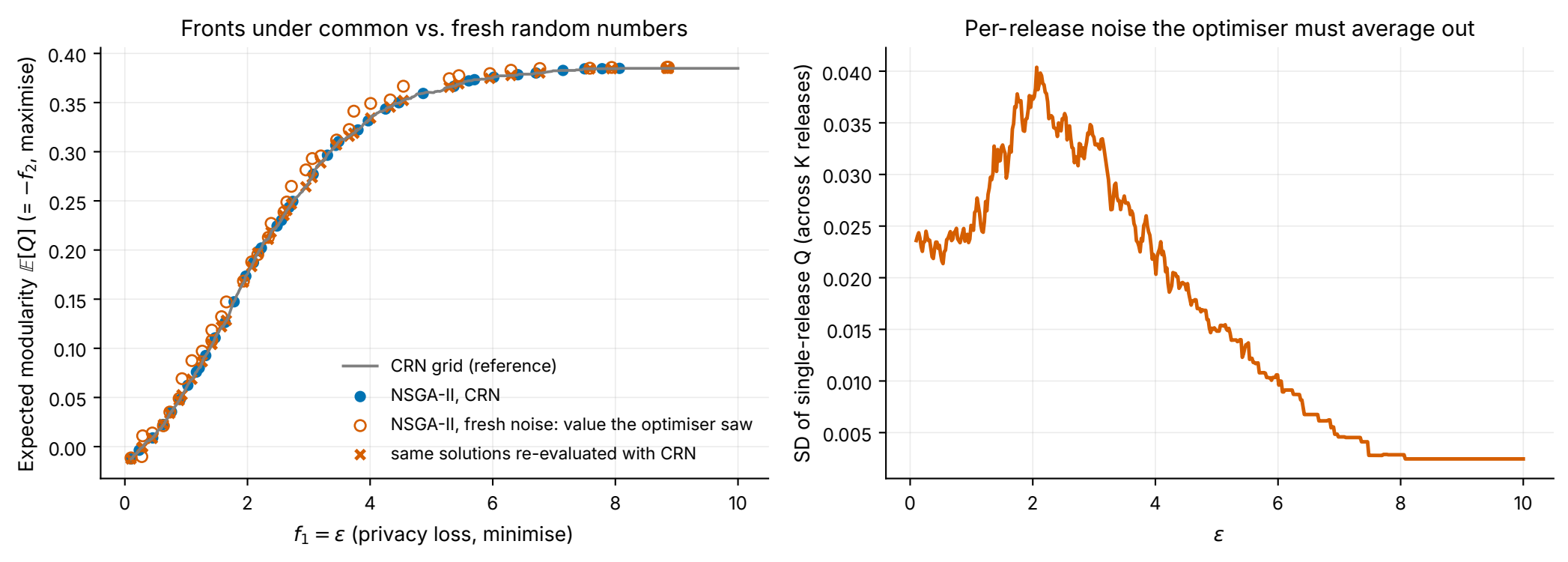
Fronts under common vs. fresh random numbers
Expected modularity $\mathbb{E}[Q]$ (= $-f_2$, maximise)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
10
$f_1 = \varepsilon$ (privacy loss, minimise)
CRN grid (reference)
NSGA-II, CRN
NSGA-II, fresh noise: value the optimiser saw
same solutions re-evaluated with CRN
Per-release noise the optimiser must average out
SD of single-release Q (across K releases)
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0
2
4
6
8
10
ε